Chokkam Mahika

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.32
SRI CHAITANYA BHASKHAR BHAVAN	Class XII (Telangana Board of Intermediate Education)	2024	96.70%
RAVINDRA BHARATHI SCHOOL	Class X (Board of Secondary Education,Telangana)	2022	10.00%

SCHOLASTIC ACHIEVEMENTS

LeetCode: Attained a **peak contest rating of 1707** (Top **15%** globally), solving **350+** Data Structures and Algorithms problems.

POSITIONS OF RESPONSIBILITY

Hospitality Coordinator | Prayas Club |

- **Spearheaded** the end-to-end planning, logistics, and execution of flagship campus events, including **Srusti Deeksha**, ensuring seamless operations and grand-scale success.
- **Collaborated** with cross-functional college student committees and management to promote events, driving high student engagement and participation.
- **Led** a team of volunteers for crowd management, venue coordination, and real-time troubleshooting during major campus initiatives.

Public Outreach Coordinator | Diesta

- Led the **end-to-end publicity**, communication strategy, and campus-wide promotion for the premier inter-hostel competition to **maximize event footprint**.
- Collaborated with **cross-functional media teams** and hostel representatives to sync marketing campaigns, boosting **student registration** and **audience turnout**.

SKILLS

Languages: Python, C++, JavaScript

Web Technologies: HTML, CSS

Core CS Concepts: Object-Oriented Programming (OOP), Database Management Systems (DBMS)

Platforms & Tools: Windows 10/11

RELEVANT COURSES

Computer Science : Introduction to Programming

Mathematics : Calculus, Linear Algebra, Differential Equations, Introduction to Probability, Introduction to Statistics, Transform Techniques, Complex Numbers.

PROJECTS

GreenRoute AI - Energy-Efficient EV Fleet Routing Simulator | **C++, OOP, Data Structures** | Github

- **Developed** a modular **C++ simulation engine** for electric vehicle fleet routing using a **custom Graph data structure** and **Object-Oriented Programming (OOP)**.
- **Implemented** an **energy-aware routing algorithm** using **Dijkstra's Algorithm** with **dynamic edge weights** based on **distance**, **traffic**, and **road elevation**.
- **Designed** modular components for **vehicles**, **packages**, **fleet dispatch**, and **file-based data management** using **STL**.
- **Simulated** vehicle dispatch, **battery consumption**, and **energy-efficient route optimization** through a terminal-based application.

Automated E-Commerce Return Fraud Risk Evaluator | *C++, OOP, Data Structures*| Github |

- Developed a real-time backend simulation to evaluate e-commerce return requests and classify them as **Instant Refund** or **Manual Fraud Inspection** using a rule-based fraud scoring engine.
- Implemented a **custom multi-level hash table** with collision handling for efficient customer record management and average **O(1)** lookup and insertion.
- Designed a **fixed-size circular sliding window** to maintain each customer's last 10 transactions, enabling constant-time updates and recent behavior analysis.
- Built a modular object-oriented architecture with separate components for transaction processing, fraud evaluation, customer management, and reporting.

Whitworth Quick Return Mechanism Simulator *(Course Team Project)* | Github |

Python | NumPy | Matplotlib | Tkinter

- Developed an interactive simulator for **4-Bar** and **Slider-Crank** mechanisms with real-time animation.
- Contributed to implementing **kinematic analysis**, including position, velocity, acceleration, and coupler curve visualization.
- Collaborated on integrating GUI controls, parameter inputs, and testing to ensure accurate mechanism simulation.

EXTRACURRICULAR

NSO Chess Player: Represented the campus in multiple tournaments, demonstrating strong strategic thinking, analytical skills, and composure under pressure.

Prayas Club Volunteer: Visited local orphanages to organize educational workshops, mentorship sessions, and recreational activities, enhancing social impact and community welfare.